

VCHECK

Vibration Checker

(for homebuilt/EAB aircraft)

DIY Open-Source Engine/Propeller Balancer - Project Status

EAA245 Presentation - August 24, 2024 (Carp, Ontario)

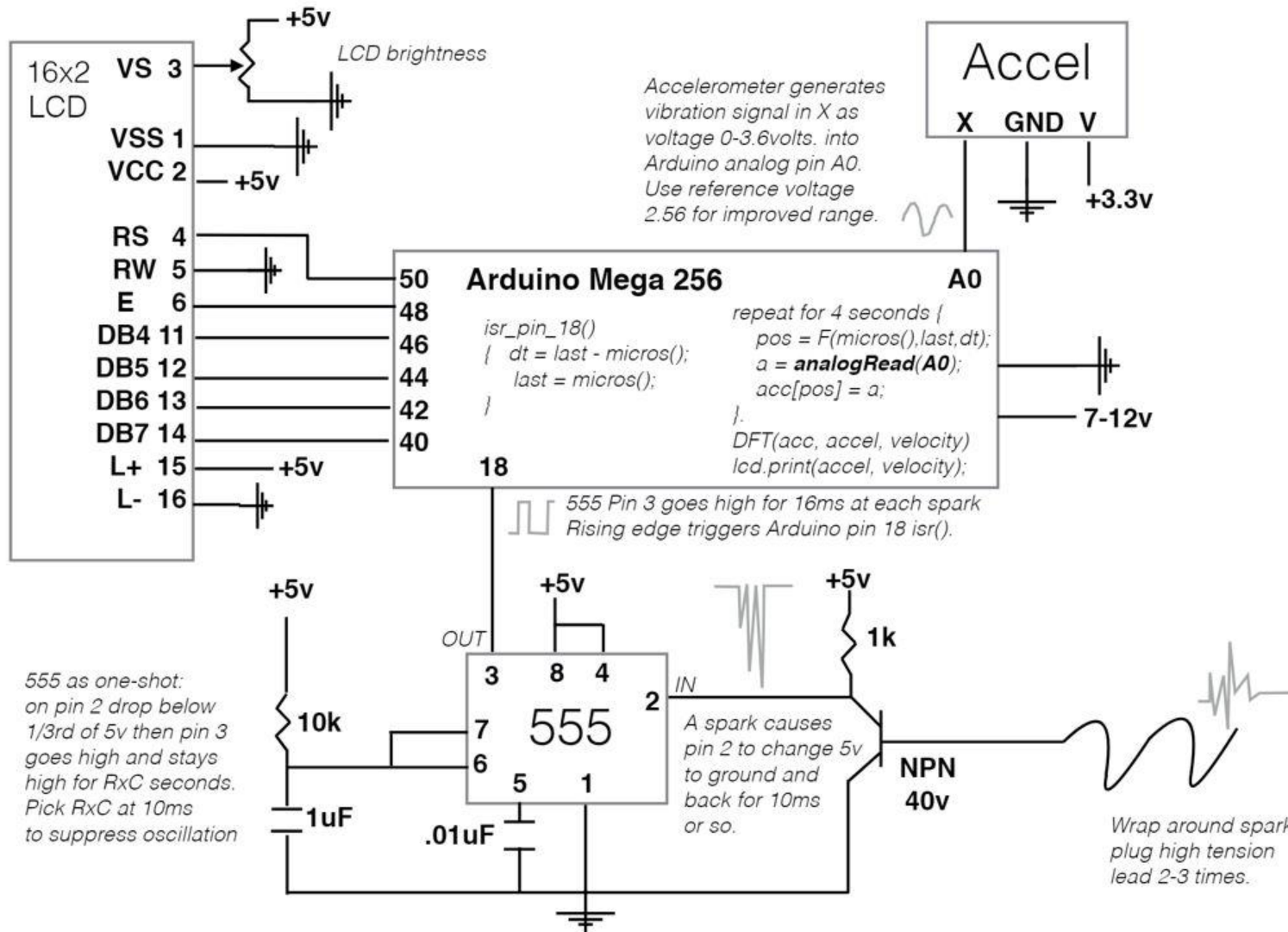
Alfio Ferrara (aka Lycosaurus)

<https://github.com/Lycosaurus>

Agenda

1. Quick Video introducing Dynamic Propeller Balancing
2. Power-point presentation
3. Short break; demonstration setup
4. Attempt 😊 at a demonstration

V-CHECK: Arduino Engine Balancer Analyzer Design



Research Pays Off

- This is my first retirement project. Wanted to re-acquaint myself with software. Little did I know ... how little I still don't know.
- Many projects on the inter webs, but this one seemed perfect. Vcheck original project by Peter Ashwood Smith (<https://github.com/peterashwoodsmith>). Searching for software/project around the globe, and I find a guy that has done this and lives just across the river from me.
- Vcheck was originally planned for capturing a trigger signal from magneto ignition leads and used an Arduino Mega 256.

First Prototypes

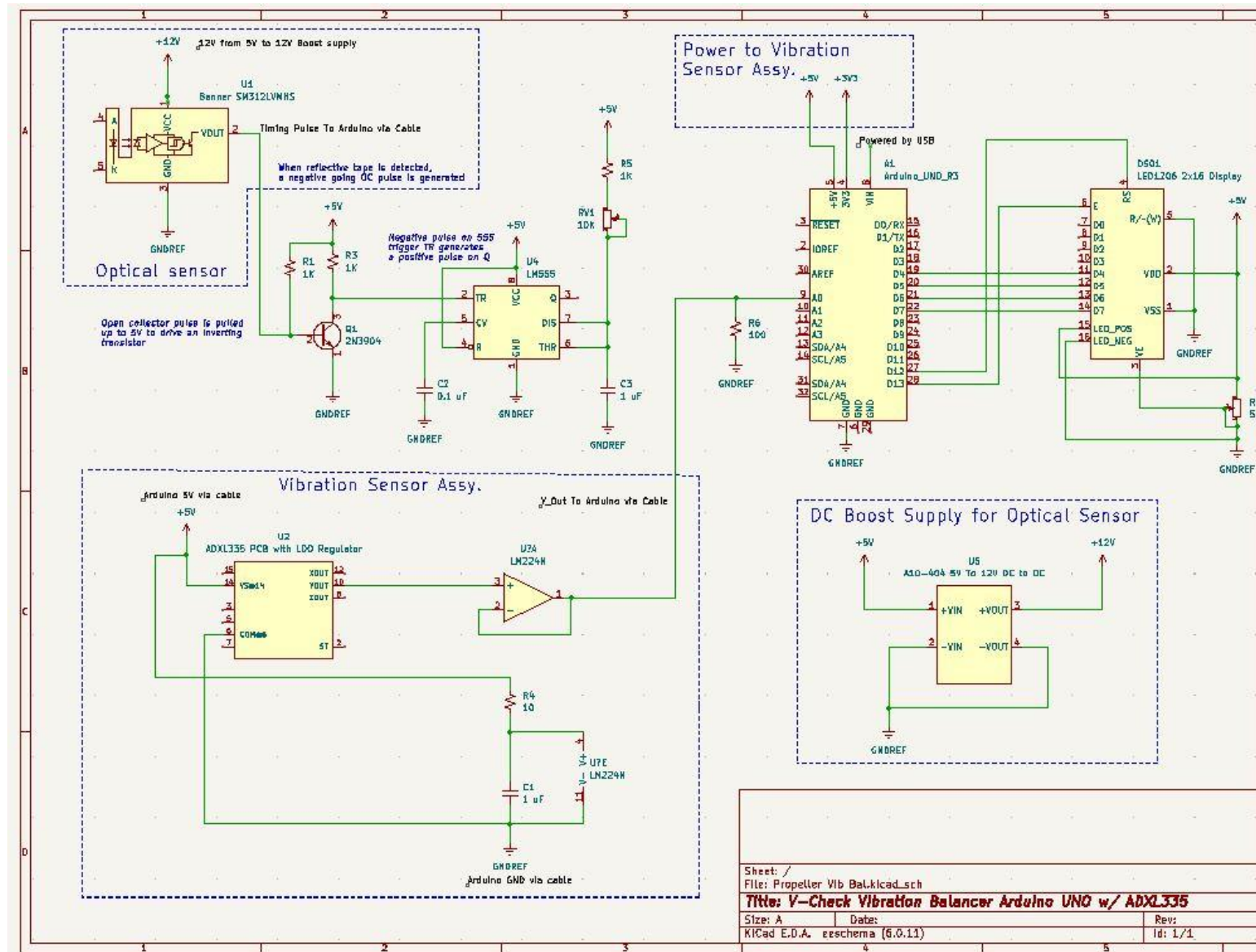
V0.1 Arduino Mega 2560 (breadboard)

V0.2 Arduino Uno (breadboard)

V1.0 Arduino Nano module-based (PCB)

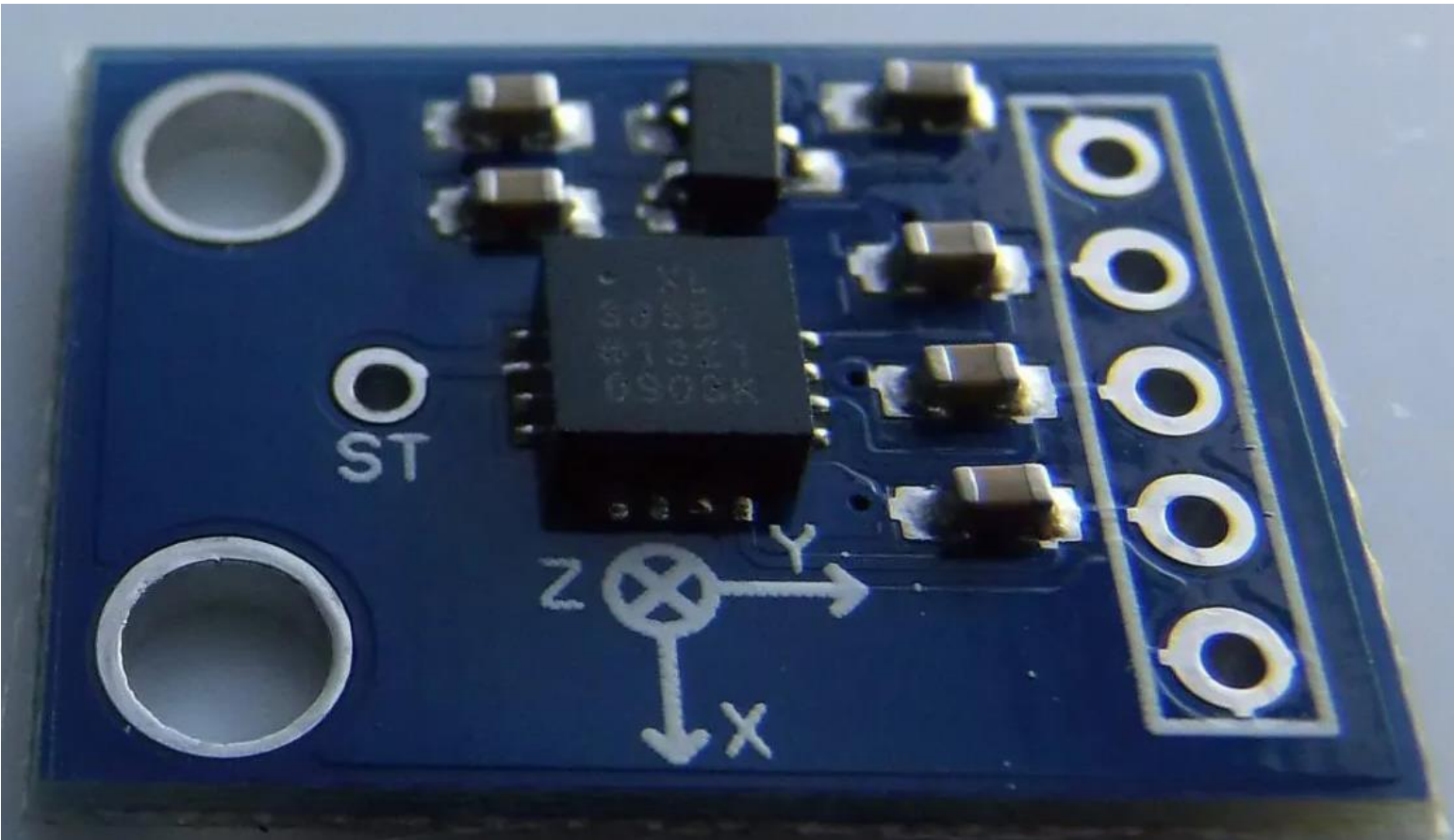
First Arduino Nano Breadboard

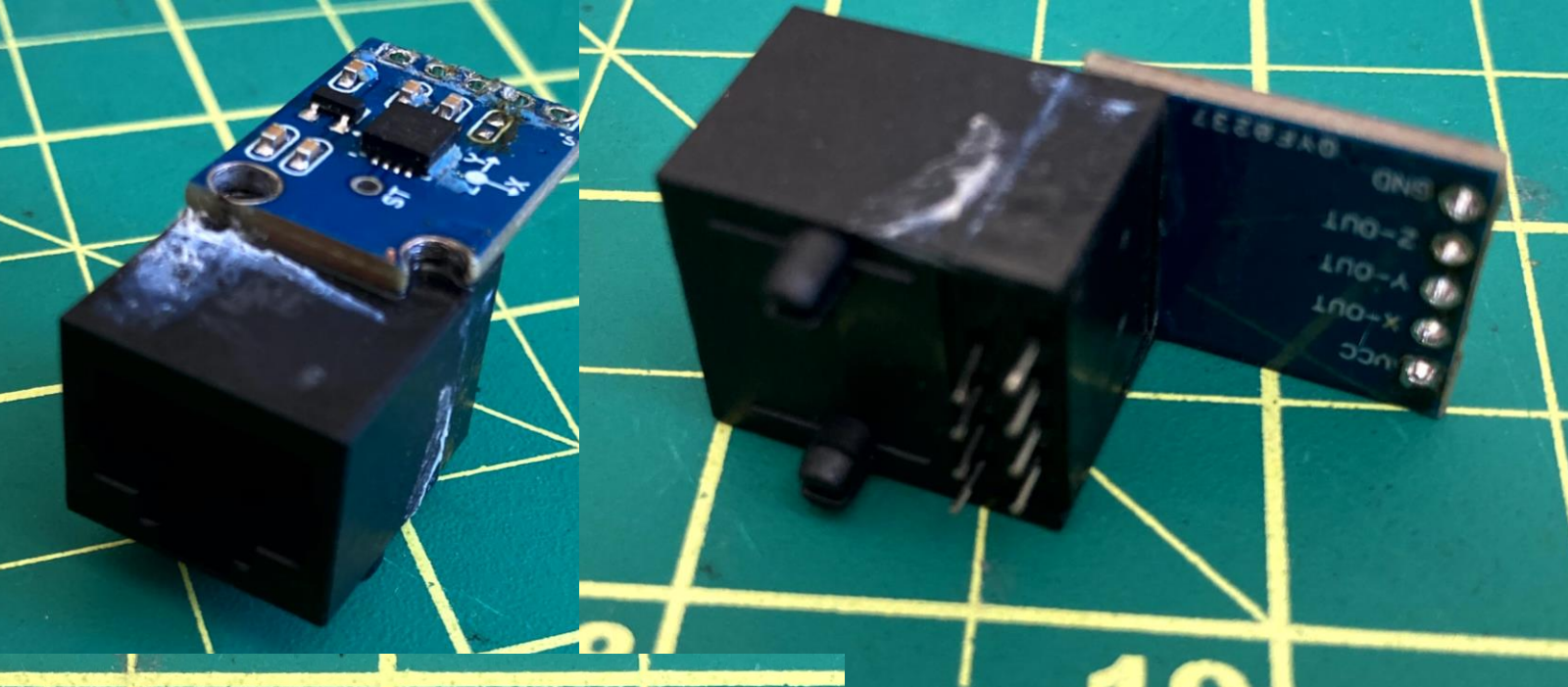
- Used a module for the switch mode (boost) supply
- ADXL335 acceleration module



ADXL335 Acceleration Sensor

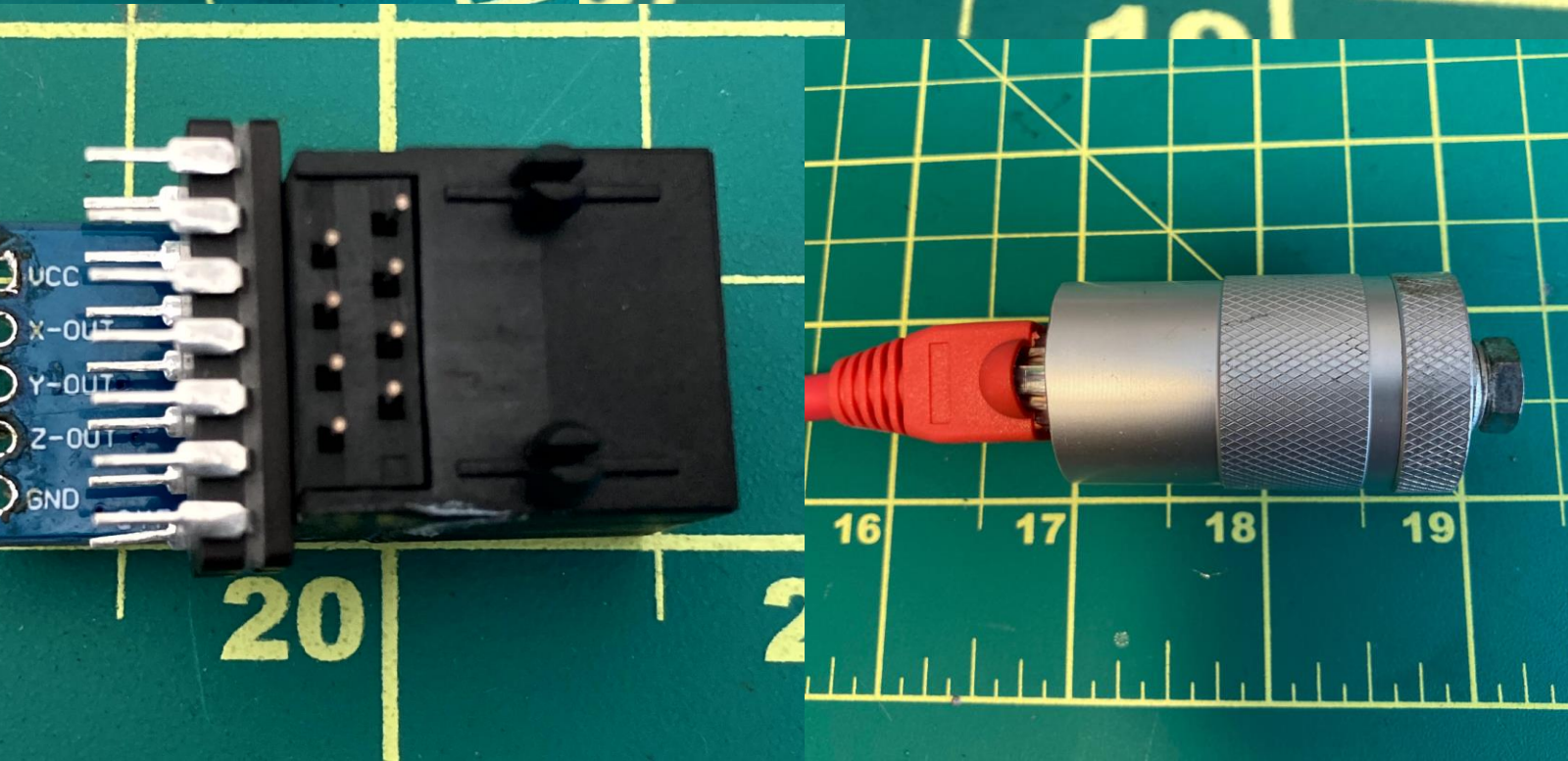
- Similar to what is already in your smartphone.
- Measures 3-axes and produces an analog signal amplitude ratiometric (proportional) to a sensed acceleration
- PCB is size of a postage stamp

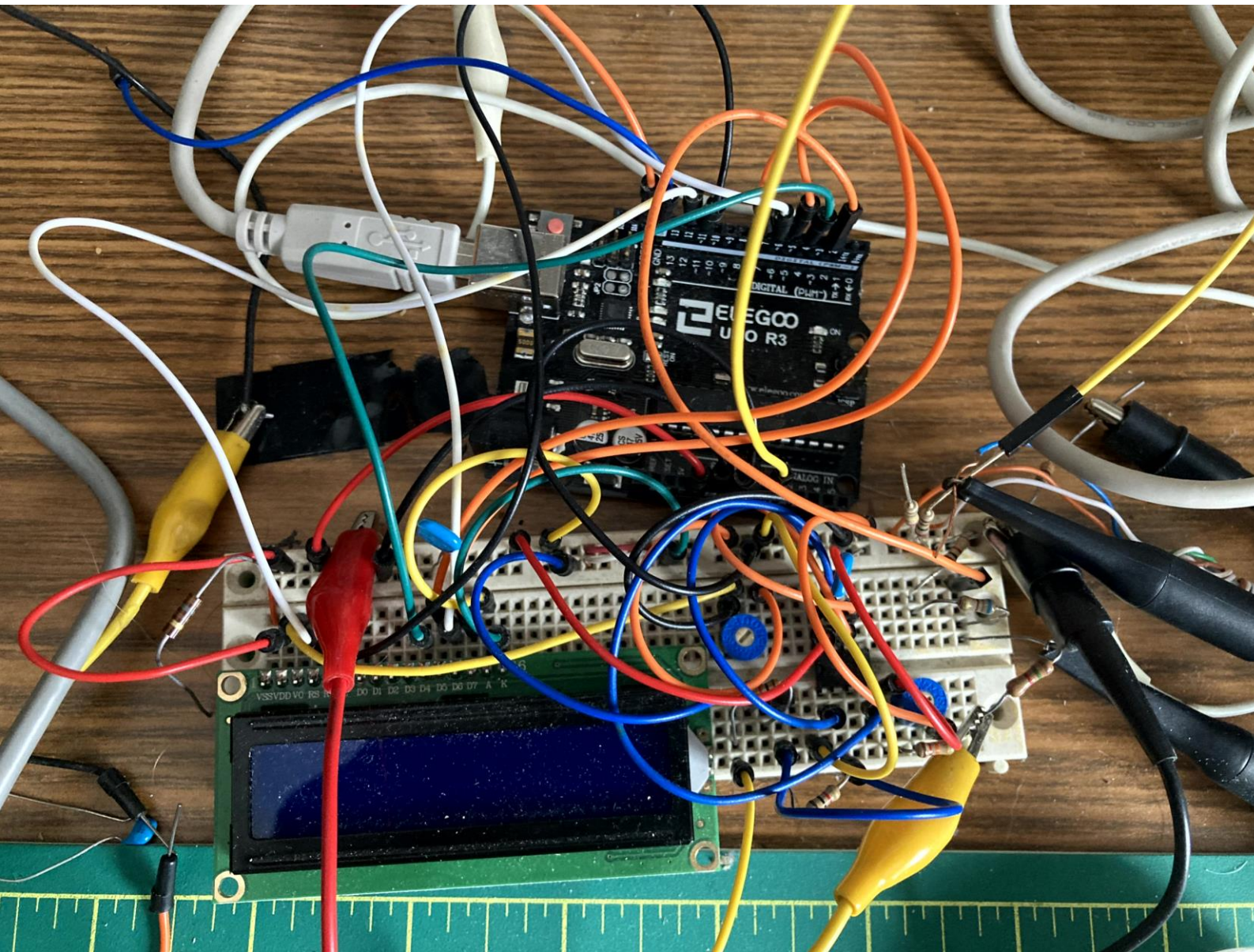




Prototype Sensor Assembly

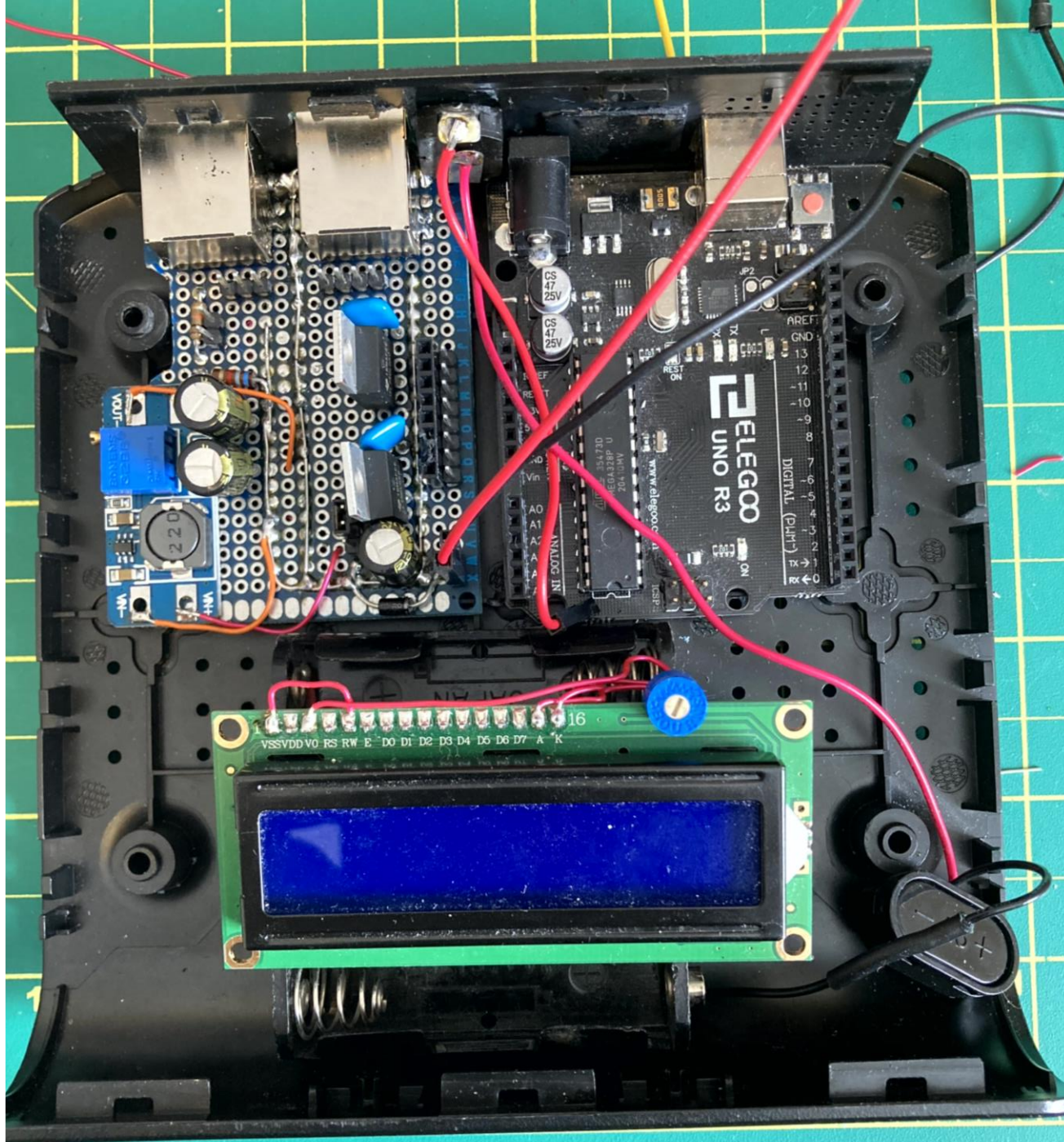
- ADXL335 sensor board glued to an RJ-45 connector, along with an op-amp
- Op-amp is used to buffer the analog signal so it can drive an ethernet cable (30 feet length tested)
- Flashlight case modified to be a sensor housing





A (messy) Prototype

- Originally assembled (not pictured) Arduino Mega 256 prototype with ADXL335 sensor and LED sync
- Attempted to use a smaller Arduino UNO R3 (pictured) graciously donated to me (Uno kit) by Cary Beazely. I mentioned to him how it would likely not have enough horsepower/ADC performance for this project ... Boy was I wrong 😊
- Messy implementation... but it worked!



Signal Processing Unit

- First integrated (into a project box) version with an Arduino Uno.
- Switched mode (boost) power supply (10.5V)
- Filtering voltage regulators (5V analog, 5V digital)
- 4xAA battery pack under 16x2 display
- Light signal conditioning, however the ADXL335 module comes with tighter filters.

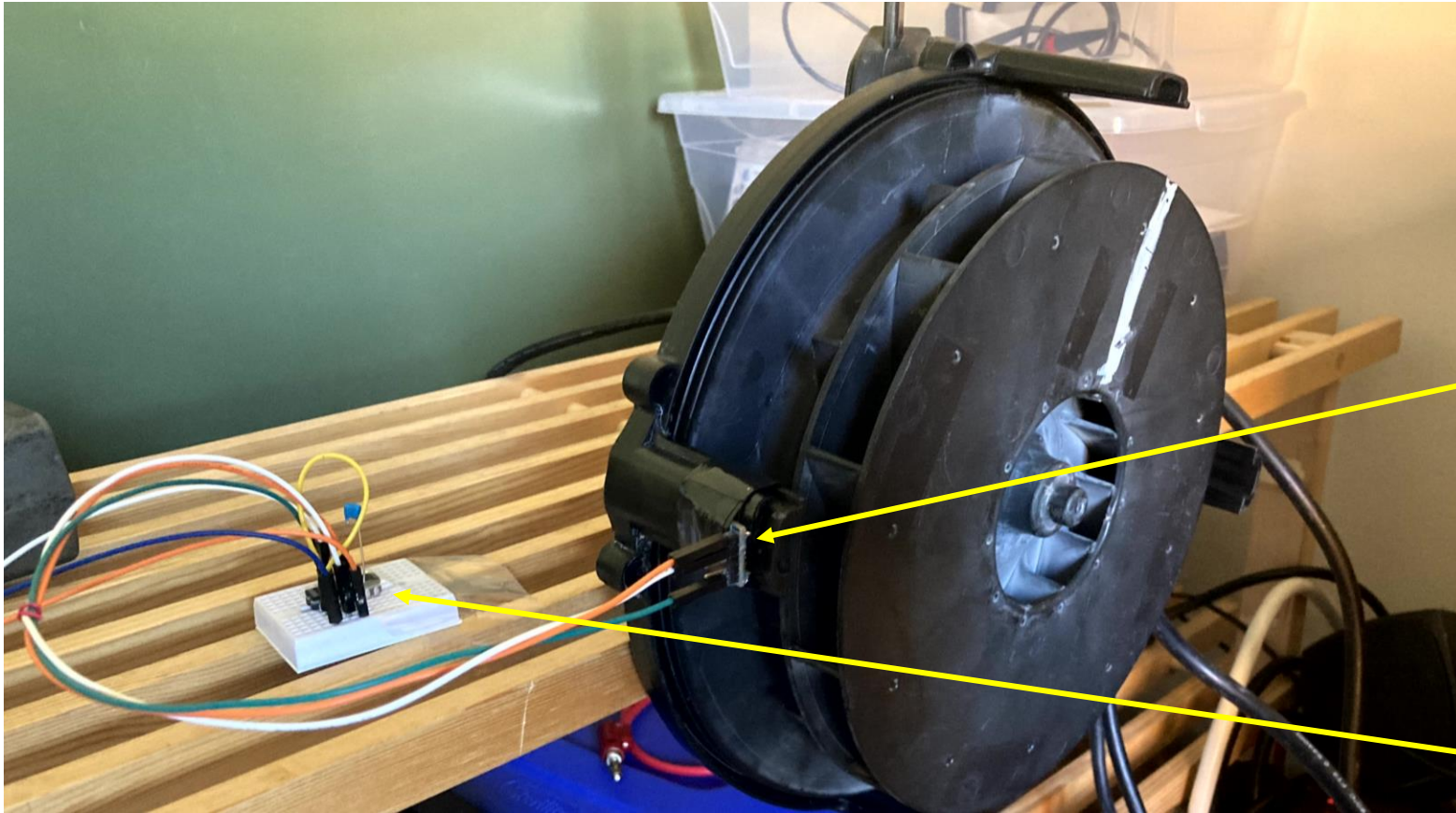
Test Setup

Fancy Record Player

- First iteration for creating a rotating imbalance
- New use for a 33LP record (reduced diameter)
- Propeller is kept for rotational mass
- 33LP not structurally sound ... bad idea



Furnace Inducer



- Another unsuccessful spinny thingy
- ADXL335 acceleration sensor
- Op-amp for driving signal onto a cable

Furnace Inducer



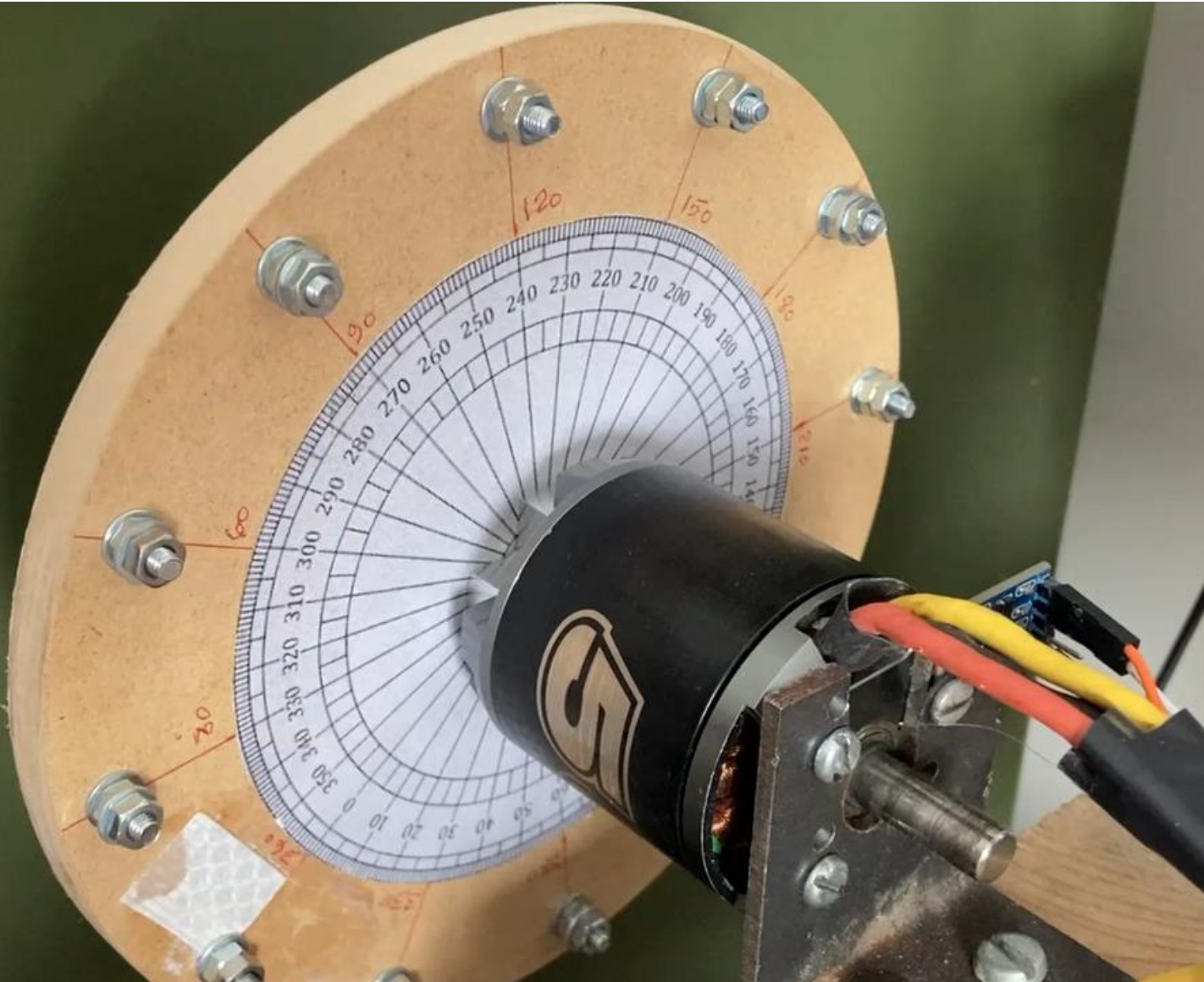
- LED sensor for sync
- Retroreflective tape
- Electrically noisy (AC powered)
- Difficult to control speed
- Using electrical tape to create off-balance... funny stories about the tape sometimes flying off at high speed

Furnace Inducer

- Inducer shown at a test RPM
- Using electrical tape to create off-balance
- Yellow trace on oscilloscope is trigger
- Blue trace on oscilloscope is acceleration signal
- Blue trace has lots of mechanical noise. Quiescent noise is actually in the hundreds of micro-Amps (measured on Vcheck SPU PCB version). FFT will do the heavy lifting of cleaning up this noise.
- Worked OK, but difficult to control speed.

Another Flywheel Thingy

- 3-phase DC powered motor system used in drones
- Lots of noise and weird vibration from motor

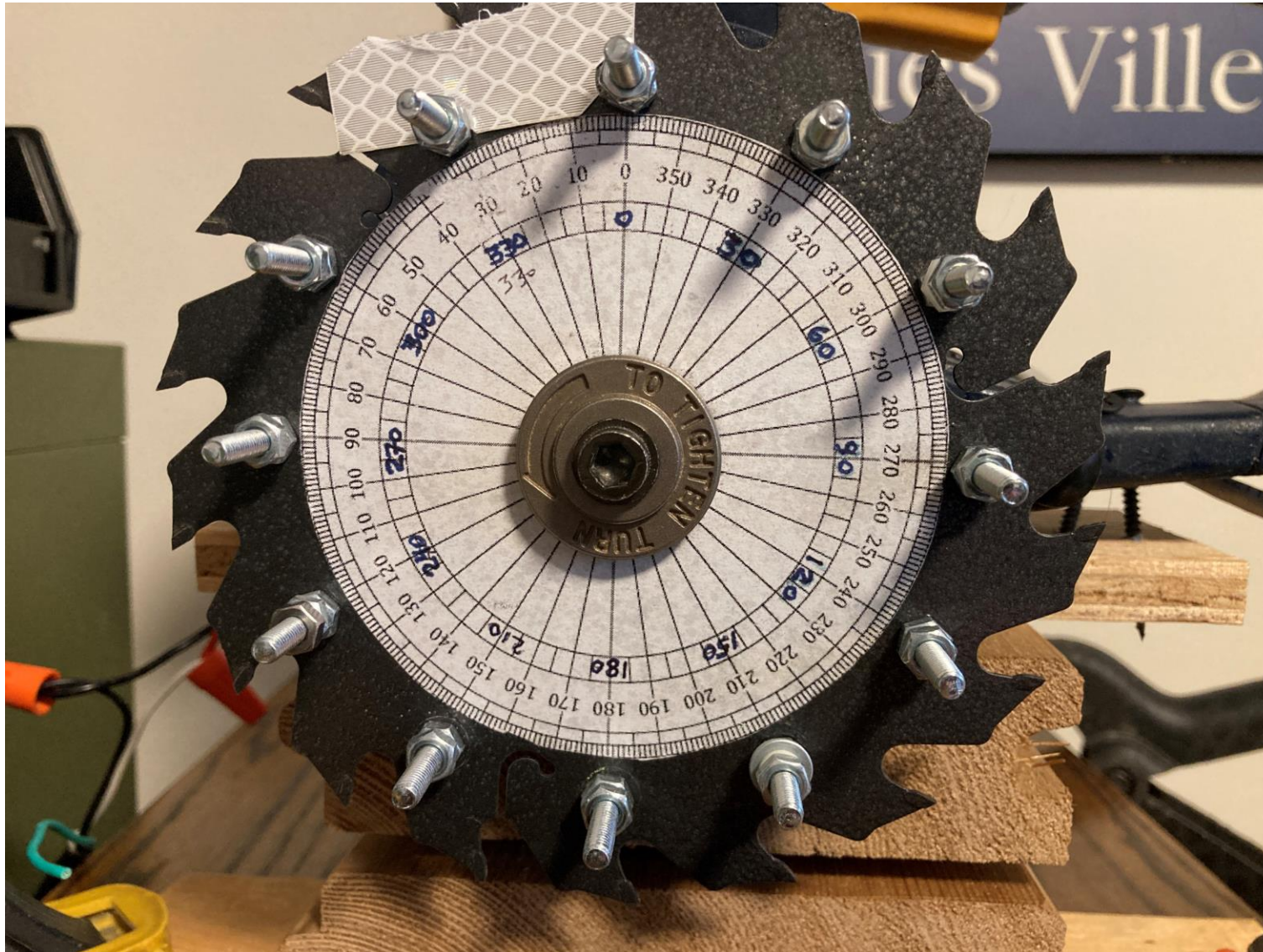




Test Whirly Thingy

(Finger Chopper 3000)

- Last Millenium portable 14.4V circular saw getting a new lease on life
- Ideal for powering via a 12-ish V variable DC supply
- Great speed control stability
- Geared, so lots of mechanical and audible noise but FFT does a good job of filtering that out



Finger Chopper 3000

- Blade reversed to “minimize” potential damage/injury
- Black paint on blade
- Drilled holes at 30 degree intervals
- Bolts/washers provide weight
- Reflector tape for sync with 0 degree “blade”
- This one’s a keeper

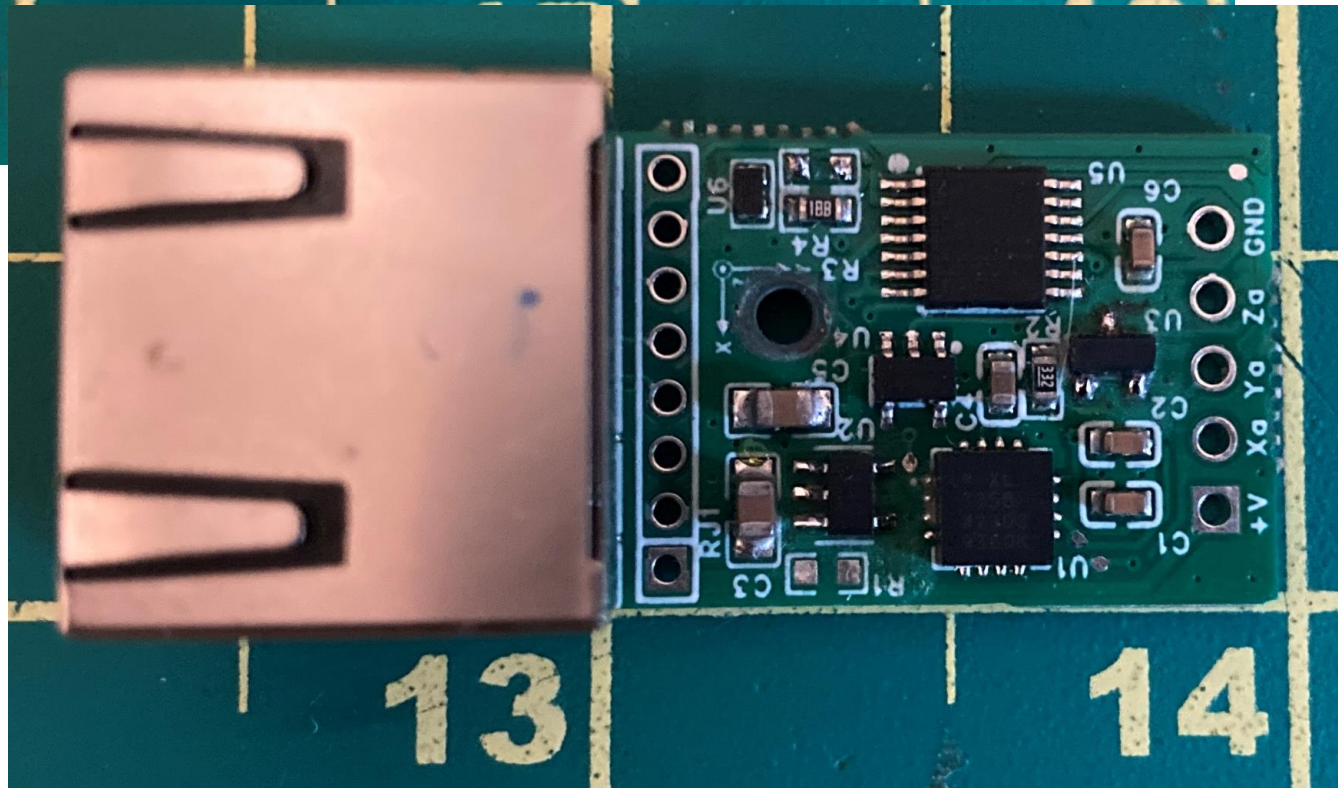
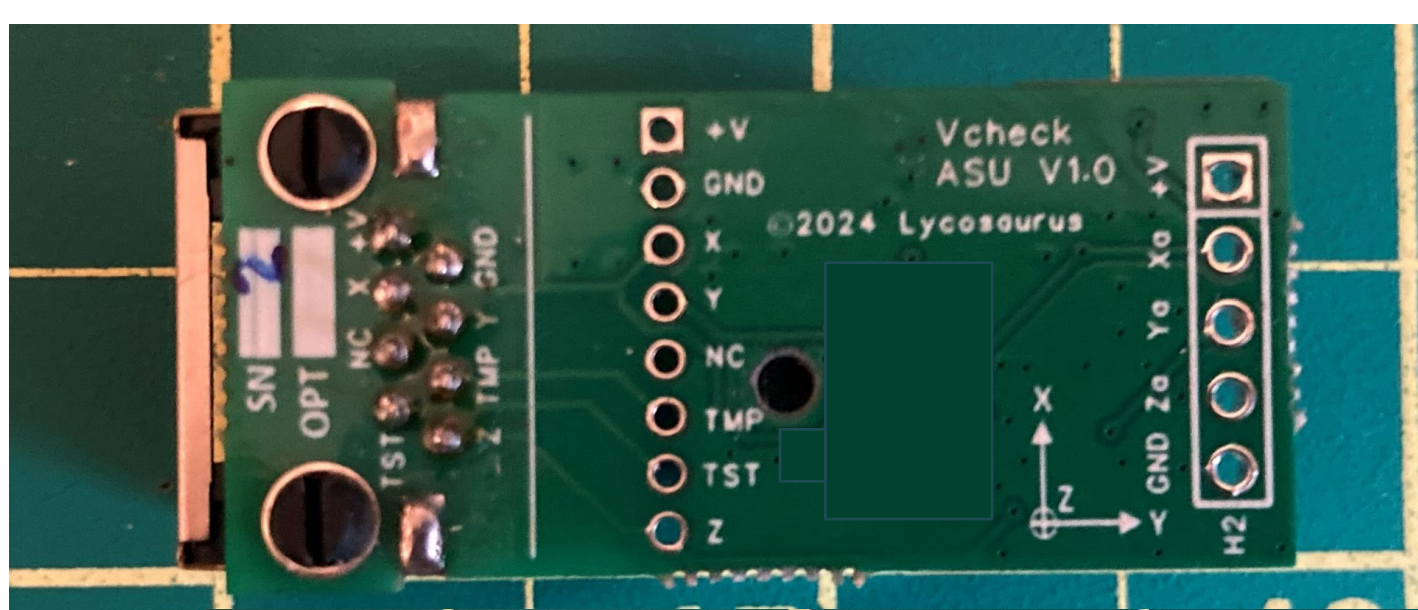
Vcheck V1.0

Features and Specifications (preliminary)

- ESD protection on critical signals
- SMPS for wide range of input voltages (4V to 20V) DC input. Can be disabled when in automotive mode (14V).
- Designed to be a final DIY product, but also highly configurable
- Low noise layout (hundreds of micro Volts)
- Low battery voltage monitoring
- Filtering voltage regulators (5V analog, 5V digital)
- Light ASU signal conditioning (better phase response over the RPM range)
- 1800 to 2700 RPM. Can go lower, needs to be tested further. Helicopters typically can use two sensors and ~500 RPM.
- Temperature monitoring capability on ASU
- Multi-axes capability. Select two axes from one or two ASU sensors.

Acceleration Sensing Unit (ASU)

- PCB design with (optional) integration of an RJ-45 connector
- RJ-45 section can be cutoff (Dremel) to make a very short sensor if wired directly
- Quad op-amp
- ESD protection for direct test line to ADXL335
- Temperature sensor
- 5V regulator (input is 5.4V)
- 3.3V regulator for ADXL335
- Optional ethernet connector
- All 3-axes (analog) sent to SPU simultaneously





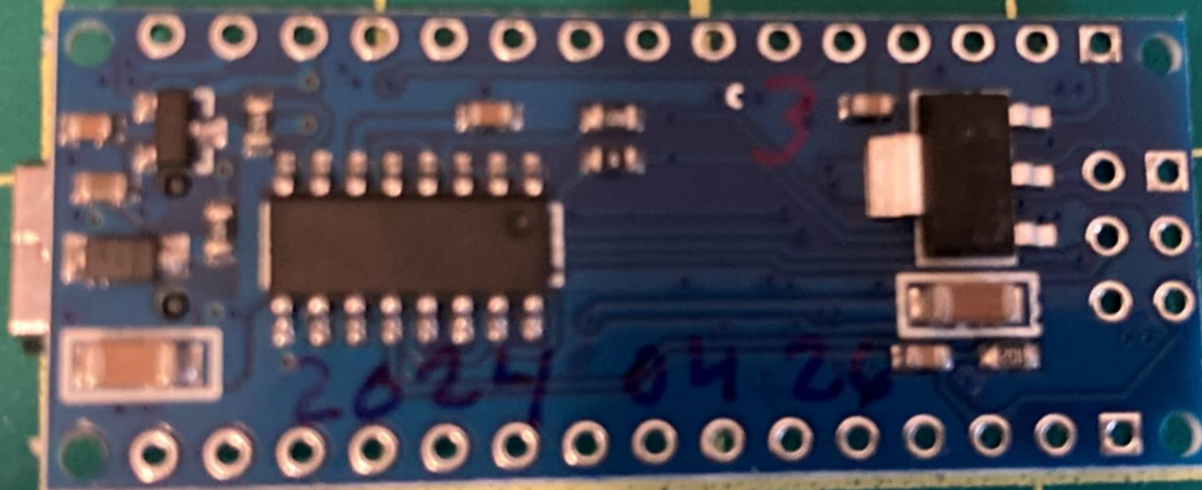
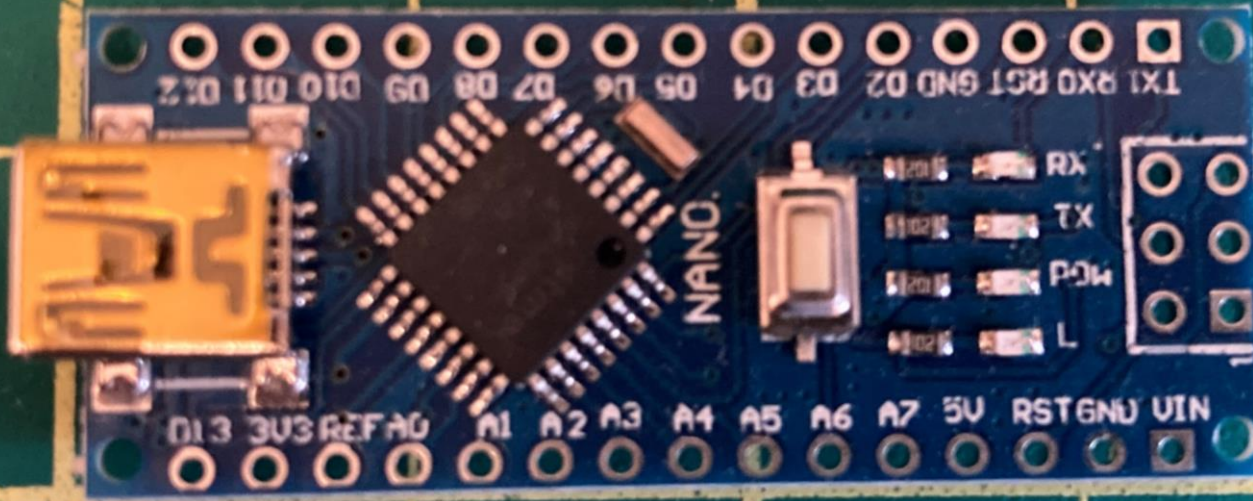
Acceleration Sensing Unit (ASU) Flashlight housing

- Takes about 30 minutes to complete one conversion (I did this one yesterday)
- Epoxy will take another 30 minutes or so to be cured enough for use
- Looks quite nice, don't you think? A couple of bucks well spent.

Signal Processing Unit

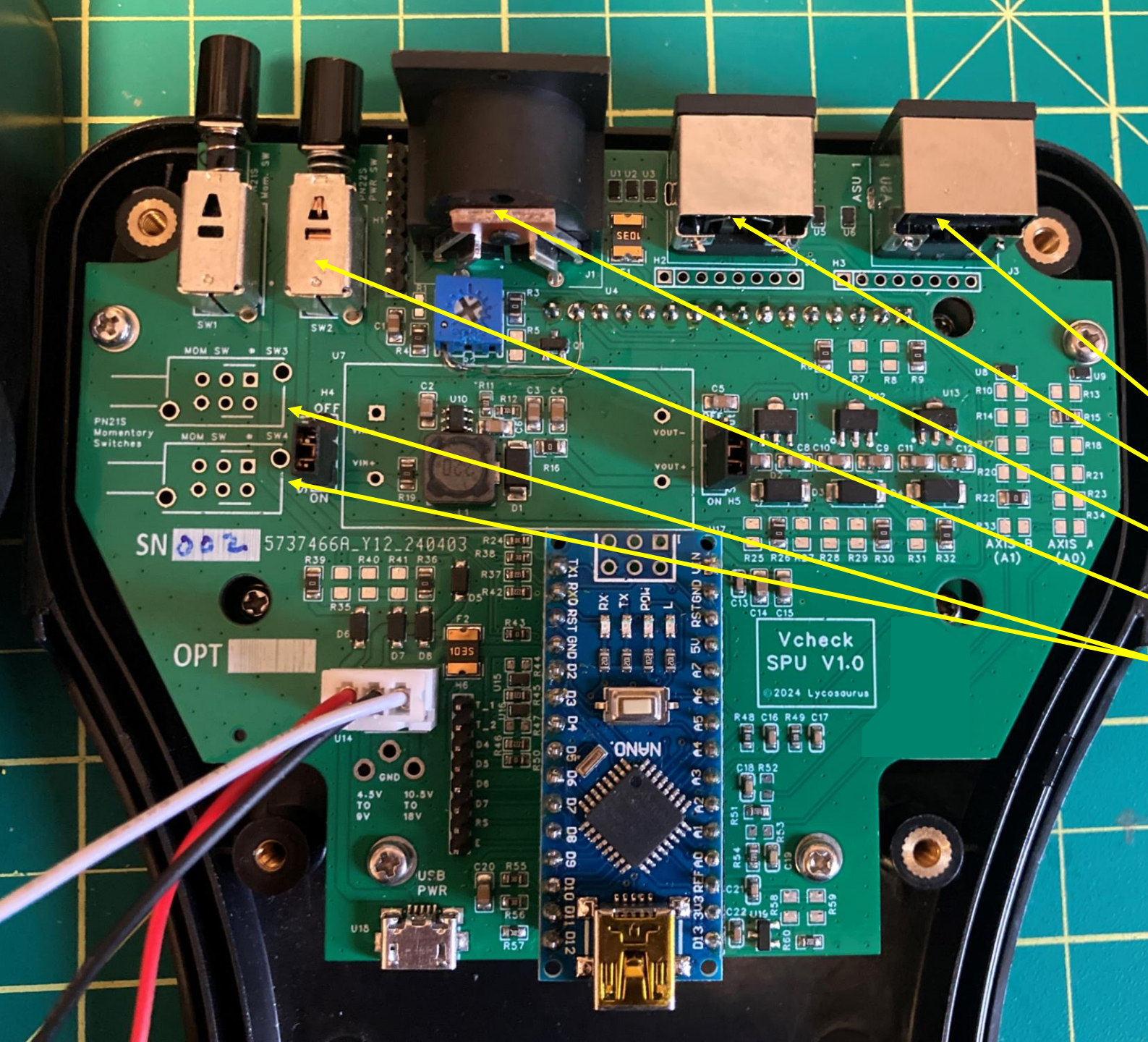
Arduino Nano

- Cost effective (\$4)
- Eases layout of Signal Processing Unit
- Allows for quick swap in case of inadvertent damage of the processor
- Size of a stick of gum

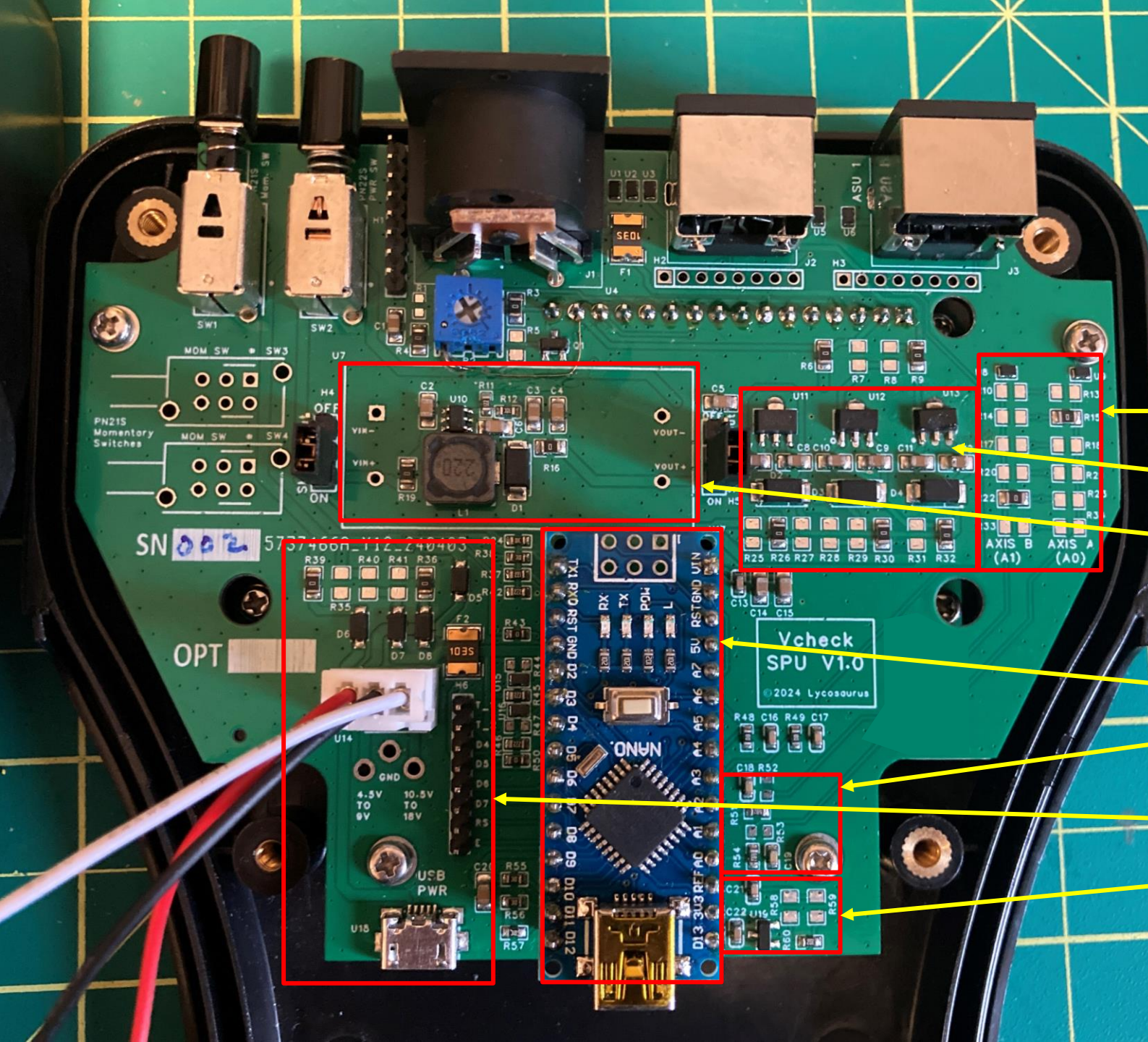


SPU

- ASU1 connector
- ASU2 connector
- Sync Pulse connector
- Input Power Switch
- Optional switches for future enhancements



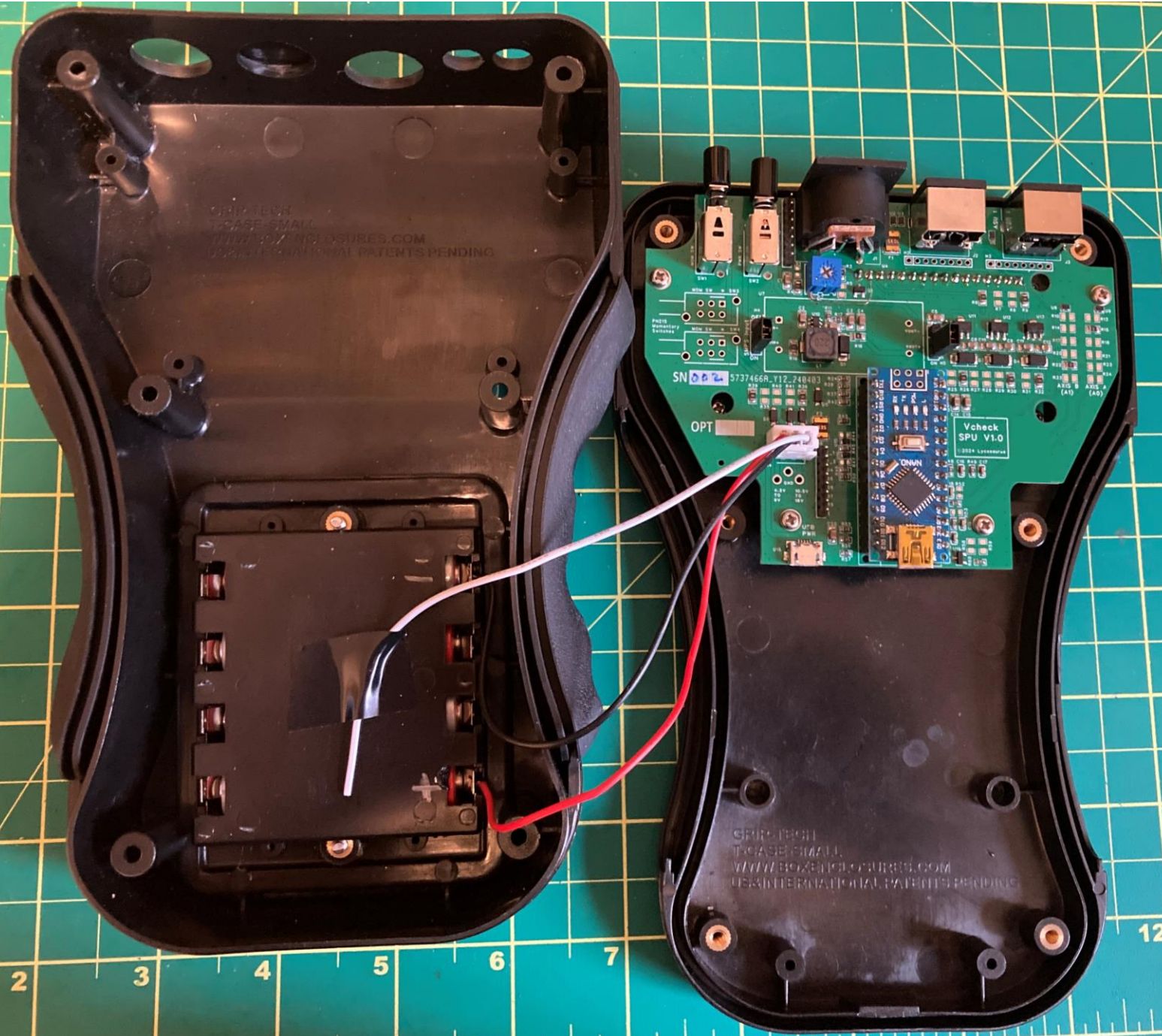
SPU

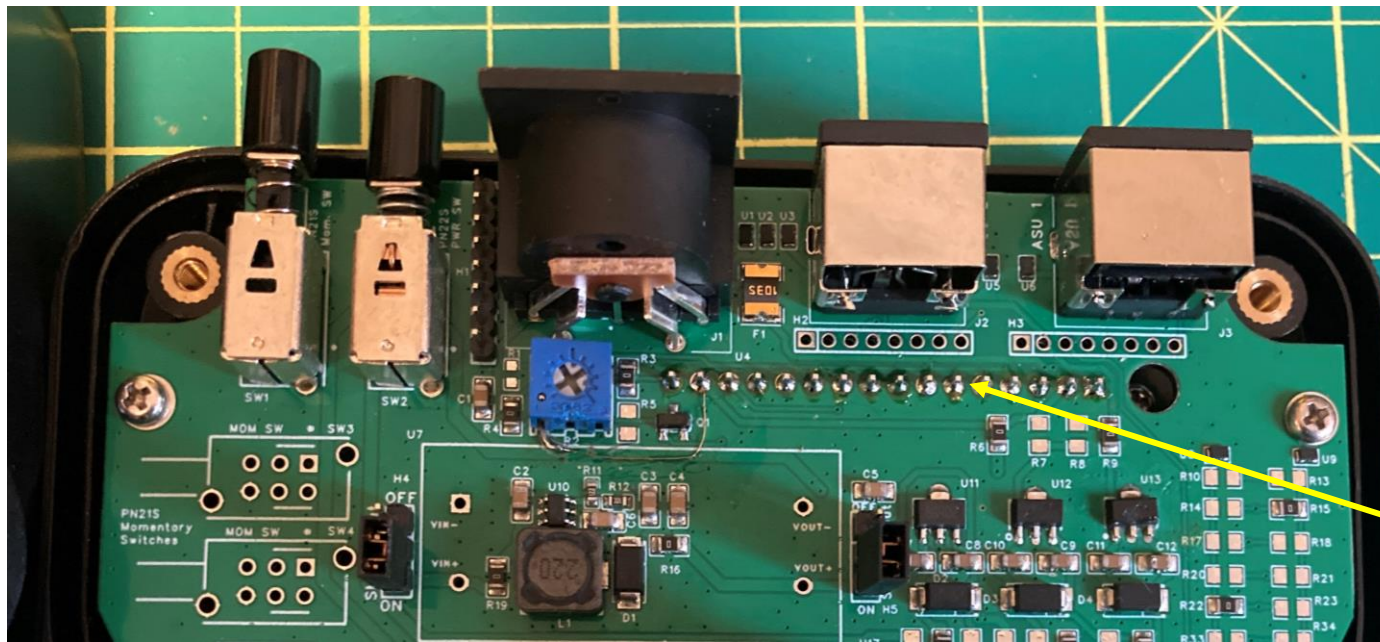


- ASU1 and ASU2 axes selection matrix
- Voltage regulators
- Switched supply (10.5V). Can use an off-the-shelf module if design did not work (derisk).
- Arduino module
- ASU acceleration signal filtering
- Input power
- Optional external voltage reference for ADC

High Quality Project Box

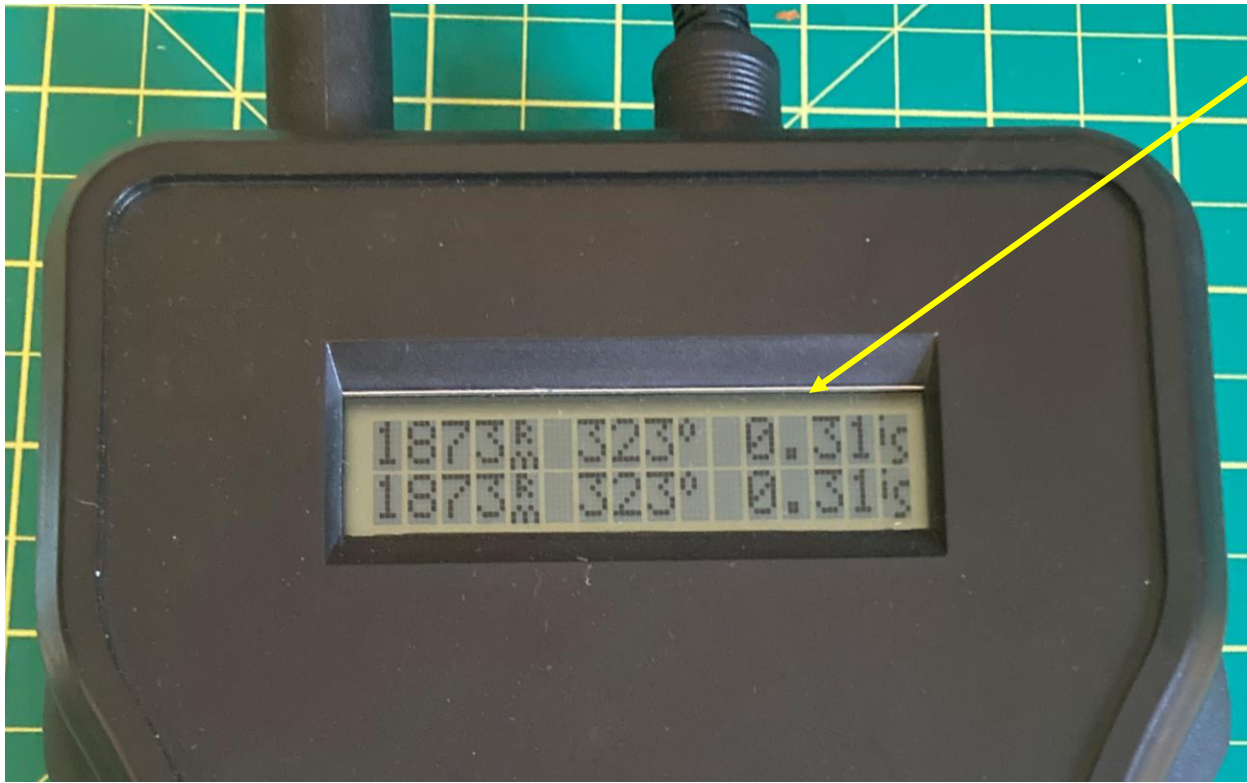
- There are not many project boxes out there that support a 16x2 LED display, along with 4xAA battery holder or single/dual 9V battery holder
- Even though the connector/switch bezel area is small, it fits (most of the) requirements. Optional switches are placed on the side, however the buttons would not be protruding from the enclosure.





Display

- 16x2 LCD display
- Mounted on solder side





Balance Weights

- Ring gear has convenient holes at 30 degree intervals
- Other options ... spinner bulkheads
- Temporary weights can be washers on spinner ... never tried it

From: Grumman Pilots Prop Bal (YouTube)

Sensor on Engine



- Sensor has an RJ-45 connector. Works well, however lacking long-term data on contact integrity with environmental conditions (humidity, contaminants), especially when under vibration
- Sensor can use an ethernet cable soldered to the sensor. Ethernet cables are cheap, especially discarded ones (check with your IT department at work).
- Retroreflective tape on propeller allows for detection even under oblique (preferred) angle of the LED beam
- Accelerometer/LED-sync support bolted onto Lycoming case bolt

Alkaline Battery Capacity Assumptions

9V : 550 mAh

4AA : 2500 mAh

Input Voltage	Current (A) (backlight OFF)	Current (A) (backlight ON)	Estimated Duration (hrs.)
6V (4AA)	.09		27
6V (4AA)		.13	19
9V	.06		4.2
9V		.08	6.8
14V (automotive)	.04		Infinite 😊
14V (automotive)		.06	Infinite 😊

Power Consumption

SMPS was ENABLED

- Current measurements from lab power supply (+/- 5 mA)
- 9V battery advantage: no leakage of an electrolyte event (that I know of). May happen but I suspect it is less of a likelihood.
- Could possibly implement a switch to swap capacity of a 2nd 9V battery for quick testing recovery (no need to open battery compartment)
- I use an external lithium battery pack (18650) to power my Arduino Uno Vcheck.
- ASU power consumption is negligible
- Note, battery voltages lower than nominal will increase current due to SMPS, thus possibly reducing duration estimate by maybe (wet finger in the wind) 10% to 15%

Item Costs (approx) \$US

SPU Signal Processing Unit	
PCB assy	15
Misc. components	8
Connectors	10
Switches	6
Arduino Nano	4
16x2 display	5
Box	40
Sub TOTAL	88

MISC	
LED Sensor	200
Ethernet Cables (free?)	20
Connectors	8
Sub TOTAL	228

ASU Acceleration Sensing Unit	
PCB ASY	15
Connector (opt)	4
Flashlight case	3
Hardware/misc	7
Sub TOTAL	29

Total Cost	345
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Cost Estimate (ballpark)

- If you want two independent sensors, add \$45 (incl. additional cable and connector)
- Note, one sensor can perform measurements in two axes. Two sensors allow measurement of two axes in different locations.
- No labour costs included... 'cause that one's free :-)
- Reflective tape can be purchased at local Harbor Freight (US) or Princess Auto (CA) (~ \$8/4ft)
- Ethernet cables are routinely chucked out into garbage/recycling by companies. Check with your IT department at work. You want the multi-stranded type. Cat-4 and above is good.
- Hope I did not miss anything... oh yeah, batteries not included.

Cost Reduction



- LED (sync) sensor could likely be cost reduced from \$200 to \$30. Some cheap knockoff sensors were tested without success, however there is a promising one I am investigating. Final system cost could go from \$345 to \$175. Stay tuned.
- Project box is really nice, however if someone has a 3D printer, a minimal feature low cost box could be constructed, supporting the present PCB size and configuration
- Project box could also be a hacked DSL/Cable modem. I built the Arduino Uno version that way box (lower left image). It's my reference unit now.



What's Next

- Re-spin the Signal Processing Unit PCB to correct some footprint errors as well as other stuff I am too embarrassed to admit to
- Investigate lower cost LED sensor implementation
- Present plan is to provide schematics etc. so people can build their own.
- Preparing an internet forum so people can share ideas.
- Test for permanent installation in a homebuilt aircraft. Monitor your vibration on-the-fly (pun intended).
- Maybe we will see Dynon/AFS/Garmin integrate this feature into their EFIS in the near future



Conclusion

- It is possible to build a dynamic vibration analyzer for a couple of hundred bucks
- Other sensors could be used, not exclusive to LED type sensor, but performance must match (response time and stability)
- Where else are you going to spend your hard earned AMUs? 😊
- IMPORTANT NOTE: You should be aware of how to use a dynamic propeller balancer, and be well aware of the inherent dangers of operating around rotating machinery and propellers. Please be smart and get appropriate training or mentoring when using this tool or any other balancing tool for that matter.